Principles of Distributed Systems

inft-3507

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Autumn 2025

Section 3: Processes, Threads and Virtualization

This content is based on the following public resources: https://www.distributed-systems.net/index.php/books/ds4/

Processes, Threads and Virtualization

Processes, Threads and Virtualization

Processes, Threads and Virtualization

Introduction to Processes and Threads

- **Definition**: A process is an independent program in execution with its own memory space, while a thread is a lightweight unit of execution within a process, sharing the process's memory.
- **Memory**: A process has its own isolated memory space (address space), whereas threads within the same process share the same memory space, including code, data, and resources.
- **Overhead**: Processes are heavier, requiring more system resources and time for creation and context switching, while threads are lighter, with lower overhead for creation and switching.
- **Communication**: Inter-process communication (IPC) is complex and slower (e.g., pipes, sockets), while threads communicate faster via shared memory but require synchronization (e.g., locks).

Processes, Threads and Virtualization

Context switching

Observations

1. Threads share the same address space. Thread context switching is much faster than process context switching:
   1. only registers and program counter need to be saved and restored
2. Process context switching is more expensive (in time and space) as
   1. TLB needs to be flushed
   2. page table is reloaded
   3. address space changes
3. Creating and destroying threads is much cheaper than doing so for processes.

Processes. Threads and Virtualization

Why use threads

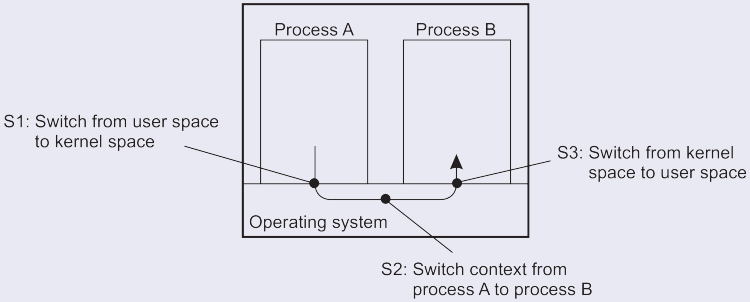
Some simple reasons

- **Avoid needless blocking**: a single-threaded process will **block** when doing I/O; in a multithreaded process, the operating system can switch the CPU to another thread in that process.
- **Exploit parallelism**: the threads in a multithreaded process can be scheduled to run in parallel on a multiprocessor or multicore processor.
- **Avoid process switching**: structure large applications not as a collection of processes, but through multiple threads.

Processes, Threads and Virtualization

Avoid process switching

Avoid expensive context switching



Trade-offs

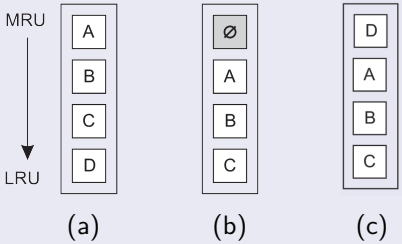
- Threads use the same address space: more prone to errors
- No support from OS/HW to protect threads using each other's memory
- Thread context switching may be faster than process context switching

Processes, Threads and Virtualization

The cost of a context switch

Suppose we have a page cache on the CPU with 4 slots. Each slot can accommodate one 4k page, associated with the instructions for a process (below, A,B,C,D)

What a context switch may cause: indirect costs



(a) before the context switch
(b) after the context switch
(c) after accessing block D.

Processes, Threads and Virtualization

Cache Perturbation Hypothesis

Ref: "The Context-Switch Overhead Inflicted by Hardware Interrupts (and the Enigma of Do-Nothing Loops)" by Dan Tsafrir (IBM T.J. Watson Research Center), presented at the ACM Workshop on Experimental Computer Science (ExpCS '07)."

Experiment

- A trivial 1 ms loop is repeatedly executed.
- Time measurements show perturbations due to interrupts.
- Some implementations showed overheads an order of magnitude larger than expected.

Hardware counters were insufficient to isolate cause. Some effects might involve:

- Instruction cache
- Data cache
- TLB
- Branch predictors

Processes, Threads and Virtualization

Thread Switch vs Process Context Switch

Thread Switch

- Switch between threads of the *same process*
- Address space remains unchanged
- Page tables and memory mappings are reused
- Faster than process switches
- Typically used for concurrency within applications

Overhead:

- Save/restore registers
- Update stack pointer and program counter

Process Context Switch

- Switch between threads of *different processes*
- Address space changes
- Page tables must be switched
- TLB often flushed or invalidated
- Slower than thread switches

Overhead:

- Save/restore registers
- Switch address space
- Update memory management state

Processes, Threads and Virtualization

Linux Kernel Threads

Threads created by the user (for example, using `pthreads`) are visible 1:1 by the kernel. If you create 10 threads using `pthreads`, the kernel sees all 10 threads.

- **Task Struct Representation**: In the Linux kernel, threads are implemented as lightweight processes, each represented by a `task_struct` (defined in `include/linux/sched.h`), sharing memory but maintaining separate execution contexts for scheduling.
- **Scheduling with CFS**: The Completely Fair Scheduler (CFS) in `kernel/sched/fair.c` manages kernel threads, treating them as virtual processors and allocating CPU time fairly using a red-black tree.
- **POSIX Threads Integration**: User-space threads (e.g., via `pthread_create`) map to kernel threads, enabling Java's 1:1 threading model to leverage Linux's scheduling for efficient concurrency.

Processes, Threads and Virtualization

Using threads at the client side

Multithreaded web client

Hiding network latencies:

- Web browser scans an incoming HTML page, and finds that **more files need to be fetched**.
- **Each file is fetched by a separate thread**, each doing a (blocking) HTTP request.
- As files come in, the browser displays them.

Multiple request-response calls to other machines (RPC)

- A client does several calls at the same time, each one by a different thread.
- It then waits until all results have been returned.
- Note: if calls are to different servers, we may have a **linear speed-up**.

Processes, Threads and Virtualization

Multithreading: does it help?

Thread-level parallelism: TLP

Let c_i denote the fraction of time that exactly i threads are being executed simultaneously.

$$TLP = \frac{\sum_{i=1}^{N} i \cdot c_i}{1 - c_0}$$

with N the maximum number of threads that (can) execute at the same time.

Practical measurements

A typical Web Server has a TLP value between 5 and 8 $\Rightarrow$ threads are primarily used for **throughput scaling**.

Processes, Threads and Virtualization

Using threads at the server side

Improve performance

- Starting a thread is cheaper than starting a new process.
- Having a single-threaded server prohibits simple scale-up to a **multiprocessor system**.
- As with clients: **hide network latency** by reacting to next request while previous one is being replied.

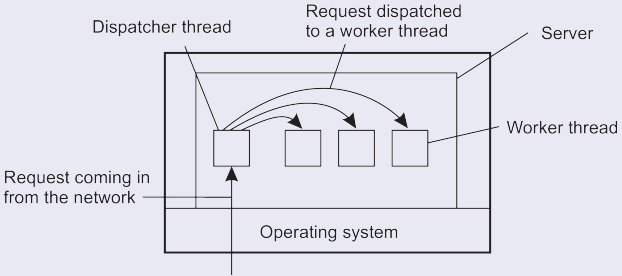
Better structure

- Most servers have high I/O demands. Using simple, **well-understood blocking calls** simplifies the structure.
- Multithreaded programs tend to be **smaller and easier to understand** due to **simplified flow of control**.

Processes. Threads and Virtualization

Multithreading - Typical Server Approach

Dispatcher/worker model



Overview

Multithreading	Concurrency, blocking system calls
Single-threaded process	No concurrency, blocking system calls

Virtualization

Goals

Goal

The primary goal of virtualization is to provide strong isolation and policy-controlled sharing of CPU, memory, and I/O resources, while preserving the expected functional and performance semantics of the software running within the virtualized environment.

Execution Virtualization Spectrum

- **Emulation** (ISA/device reproduction): maximizes compatibility across architectures, typically higher overhead
- **Process VM** (language/bytecode runtime): portability and safety for applications, not a full OS boundary
- **System VM** (hypervisor-based): strong isolation and full OS virtualization

Key distinction

Process virtual machines virtualize the **execution environment** of a **single program**, whereas hypervisors virtualize the underlying hardware interface, enabling **entire operating systems** to run as if on a physical machine.

Taxonomy of Virtualization Approaches

Emulation
- Virtualizes an ISA and/or devices in software
- Primary benefit: cross-architecture compatibility

Process Virtual Machine (e.g., JVM)
- Virtualizes an application execution model (bytecode + runtime services)
- Primary benefit: portability, safety properties, managed services

program boundary

System Virtual Machine (Hypervisor / VMM)
- Virtualizes CPU, memory, and devices to run an unmodified OS
- Primary benefit: strong isolation and resource control at machine granularity

OS boundary

Figure: Conceptual taxonomy. Emulation targets compatibility, process VMs target portable execution for programs, and system VMs target full OS virtualization and isolation.

System VMs: Type 1 and Type 2 Hypervisors

- **Type 1 (bare-metal):** hypervisor runs directly on hardware, commonly with a privileged management domain
- **Type 2 (hosted):** hypervisor/VMM runs atop a general-purpose host OS and leverages host drivers/services

Engineering focus

The practical security boundary is determined by the **trusted computing base (TCB)**: what must be trusted for isolation.

Type 1 vs Type 2 Stack Organization

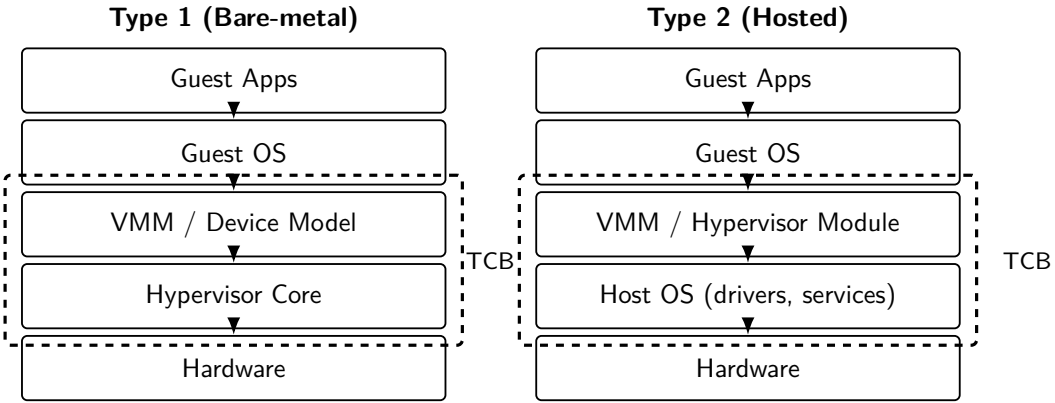


Figure: Representative stack organizations. In Type 2 systems, the host OS becomes part of the trusted base for isolation.

Emulation vs Hypervisor-Based Virtualization

- **Emulation:** can execute software for a different ISA; overhead arises from instruction translation and device modeling
- **System VM with hardware assist:** guest code runs largely natively; overhead concentrates in traps/exits and I/O paths
- **Process VM (e.g., JVM):** does not virtualize devices or privileged CPU state; relies on the host OS for isolation

Implication

ISA portability and OS-level isolation are different objectives; they are frequently conflated but have different mechanisms and costs.

Virtualization in Cloud Computing

Virtualization is the foundational technology that enables modern cloud platforms.

In cloud computing, virtualization:

- Allows multiple **virtual machines (VMs)** to run on a single physical server
- Provides strong **isolation** between tenants for security and fault containment
- Enables **elastic resource allocation** (CPU, memory, storage) on demand
- Makes infrastructure **hardware-agnostic**, simplifying scaling and migration

Cloud providers rely on virtualization to offer Infrastructure-as-a-Service (IaaS), where users deploy full operating systems without managing physical hardware.

Resource Controls

Resource Control: CPU and Memory

Resource Control refers to mechanisms used to **allocate** and **limit** finite resources according to defined policies. The goal is **predictable performance**, **isolation**, and **stability** across competing workloads.

CPU control concepts

- **Capacity:** number of vCPUs and their mapping to physical CPUs
- **Shares/weights:** proportional allocation under contention
- **Limits/caps:** enforce maximum CPU usage (often expressed as a percentage of a core set)
- **Reservations:** minimum guaranteed CPU capacity (where supported)

Control groups (`cgroups`): accounting and limits

cgroups group processes and apply resource *accounting, limits, and prioritization*.

Typical controls:

- **CPU**: shares/quotas; prevent CPU hogging
- **Memory**: limits; OOM behavior per group
- **I/O**: throttle block I/O
- **PIDs**: limit number of processes

Key idea: **namespaces isolate**; **cgroups constrain**.

Control Group Examples (CPU % and Memory %)

CPU

- **CPU cap (percentage):** limit a VM to (e.g.) 60% of one core or 240% of a 4-core entitlement
- **Shares/weights:** if two VMs have weights 2:1, they receive that proportion under contention
- **Pinning/affinity:** restrict vCPUs to selected cores to control interference and improve predictability

Memory

- **Memory limit:** hard upper bound on guest-usable memory
- **Reservation:** minimum memory intended to remain available (platform-dependent)
- **Overcommit with ballooning:** reclaim unused guest pages before swapping at the host

cgroups as a resource governance tree

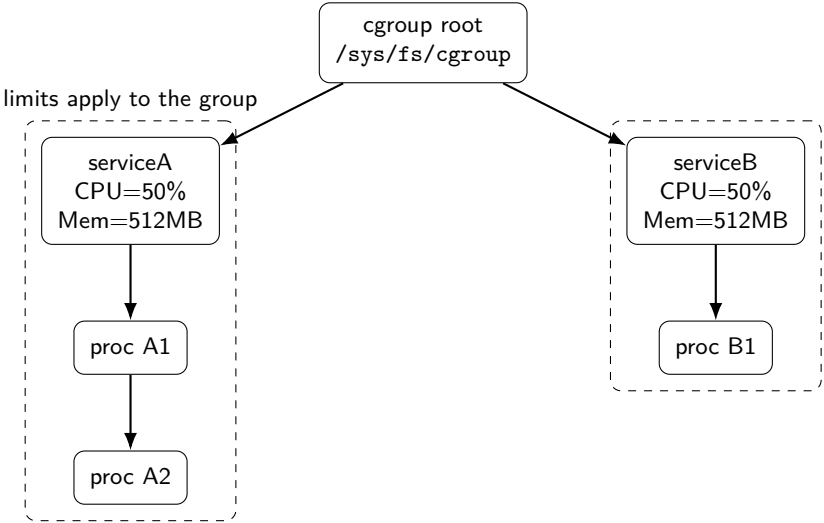


Figure: Cgroups organize processes into a hierarchy and apply resource limits/accounting per group.

Namespaces: isolation of kernel resources

A **namespace** gives a process a private view of a specific kernel resource.

Common namespaces:

- **PID** (process IDs): processes see a different PID tree
- **Mount** (MNT): separate mount table / filesystem view
- **Network** (NET): interfaces, routes, ports isolated
- **UTS**: hostname/domainname isolation
- **IPC**: SysV IPC, POSIX message queues, etc.
- **User** (USER): user/group ID mapping (enables unprivileged containers)
- **Cgroup** namespace: hides cgroup paths

Namespaces provide separate “views”

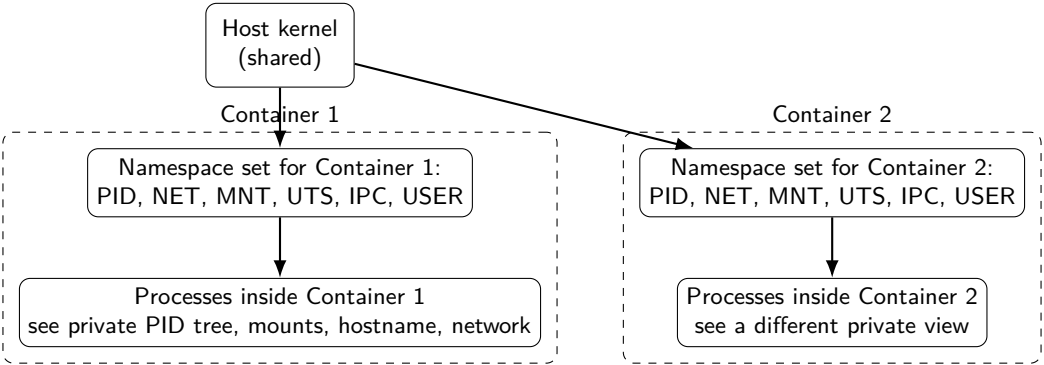


Figure: Namespaces isolate what processes can see; multiple containers can share one kernel while keeping distinct views.

A simple isolation example with `unshare`

`unshare` creates new namespaces for a process (and its children).

This demonstrate isolation by creating:

- a new **UTS namespace** (private hostname)
- a new **PID namespace** (PID 1 inside)
- a new **mount namespace** (private mount table)
- optionally a **user namespace** (run as "root" inside without host root)

How they fit together (container model)

A container runtime typically:

1. Creates new **namespaces** for the process (isolation)
2. Places the process into **cgroups** (resource governance)
3. Sets up a root filesystem and mounts (often with overlayfs)
4. Configures networking (veth pairs, bridges, NAT, etc.)

The result: a process that *looks* like it runs on its own system and can be *limited*.

Example: unshare for a private hostname + PID tree

```
# Create new user, mount, UTS, and PID namespaces.
# --map-root-user maps your user to root inside the user namespace.
unshare --user --map-root-user --mount --uts --pid --fork /bin/bash
```

Inside the new shell:

```
# UTS namespace: hostname is private
hostname container-demo

# PID namespace: this shell's child processes start a new PID tree
echo "my pid:" $$

# Compare process listing (you should see a tiny PID universe)
ps -ef
```

Example: mount namespace isolation (private mounts)

In the same `unshare` shell:

```
# Make mount operations "private" to this namespace
mount --make-rprivate /

# Create a temporary mount only visible here
mkdir -p /tmp/ns-mnt
mount -t tmpfs tmpfs /tmp/ns-mnt

# Observe it exists here:
mount | grep ns-mnt
```

On the host (outside the namespace), that tmpfs mount will not appear in `mount` output.

The Isolation Achieved

From the example:

- **UTS namespace**: hostname change does not affect the host
- **PID namespace**: processes see a new PID hierarchy (often with PID 1 inside)
- **Mount namespace**: mounts are private to the namespace
- **User namespace**: you can have root-like IDs inside without being host root

This is the core of container isolation, and it ties directly back to virtualization:
virtualize the process's view of the system without virtualizing the hardware.

Containers

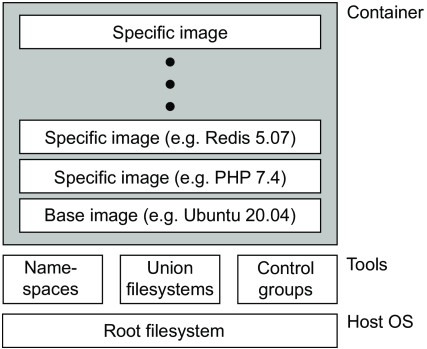
Containerization in Cloud Computing

In cloud environments, containers:

- Package applications with their dependencies for **portable deployment**
- Share the host kernel, enabling **higher density** than VMs
- Start quickly, supporting **rapid scaling** and auto-healing services
- Integrate naturally with orchestration systems (e.g., Kubernetes)

A **container** is typically: *namespaces* + *cgroups* (plus filesystem + tooling). As a result, containerization is central to Platform-as-a-Service (PaaS) and microservice-based architectures, enabling fast iteration and efficient use of cloud resources.

Containers



If multiple applications share one OS kernel, we want:

- Strong **separation of views** (processes, network, mounts, hostnames, users)
- Predictable **resource sharing** (CPU, memory, I/O) without one app starving others

The **Union file system:** combine several file systems into a layered fashion with only the highest layer allowing for `write` operations (and the one being part of a container).

Summary

Summary and Conclusions

We have discussed the following topics in this part of the course, namely:

- Processes and Threads
- Context Switching
- Multithreading
- Virtualization
- Resource Management and Isolaiton
- Containerization